7 Empirical Models

Empirical. *In matters of art or practice: That is guided by mere experience, without scientific knowledge.*
—Oxford English Dictionary, 2002 [Dic02]

7.1 Overview

Much of the book so far has focused on fundamental models of nervous system dynamics. But the control framework we have discussed is applicable regardless of the derivation of the model.

Using fundamental models assumes that you know what you're doing. It assumes that you know so much about the system that you really can write down from first principles models of neurons, models of connections for those neurons and the functions of their synapses, and models of the ions around them, with a better end result than you would get from guessing and, in a sense, better than what you could do from analyzing the data itself without the model. Only in the past few decades can we feel justified in the hubris that the previous sentence might be true.

In this chapter, on the other hand, we discuss models that assume your prior knowledge is really inadequate for observing and controlling neuronal systems. I focus on a class of decompositional techniques that often flexibly described in the literature as principal component analysis, singular value decomposition, Karhunen-Loève decomposition, and principal orthogonal decomposition. The underlying fundamentals of these techniques are similar, and I focus on those fundamentals here.

We consider a matrix, A, acting on a vector, x, as Ax. The matrix A is a set of scalar coefficients that will transform the vector by changing the coordinates that the vector components represent. These changed coordinates take the vector from one vector space, the domain, to another vector space, the range. The transformation can remain in the original space, but often (the focus of this chapter) the transformation is to a different and smaller space.

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Singular value decomposition is a matrix decompositional technique. One starts with any matrix A.[1] A square matrix is symmetric if

$$A^T = A$$

where T indicates transpose. If i and j are rows and columns, then $A_{ij}^T = A_{ji}$. For any *symmetric* $m \times m$ matrix A with real entries, there is an orthonormal eigenbasis, a set of real-valued eigenvectors[2] $w_1, \ldots, w_m$. The orthonormality says that $w_i^T w_j = 0$ if $i \neq j$, and otherwise $w_i^T w_i = 1$. For any A, not necessarily symmetric, $A^T A$ is symmetric.

In previous chapters, we dealt with covariance matrices in Kalman filtering. If we start with a matrix of data values A, where each row represents a different sample in time, and each column represents a different sensor or electrode, then forming $A^T A$, or AA^T, will turn this matrix into a square symmetric covariance matrix.[3] Recalling that $(AB)^T = B^T A^T$, then to show that $A^T A$ is symmetric (even if A is not symmetric) we can write

$$(A^T A)^T = A^T (A^T)^T = A^T A$$

Furthermore, the eigenvalues of $A^T A$ must be nonnegative. In fact, if v is a unit eigenvector, and $Av = \lambda v$, then

$$0 \leq |Av|^2 = v^T A^T A v = \lambda v^T v = \lambda$$

There are then two ways to make square symmetric correlation matrices from typically rectangular n x m data matrices A: $A^T A$ and AA^T. Each has an orthonormal set of eigenvectors. One set of eigenvectors comes from the rows of A, and one comes from the columns. These rows and columns span different spaces, in general, so we will distinguish the set of m unit eigenvectors of $A^T A$ as $v_1, \ldots, v_m$, and the n unit eigenvectors of AA^T as $u_1, \ldots, u_n$. We are going to use the left and right multiplied matrices throughout this chapter to create these different covariance matrices.

Singular value decomposition computes these two orthonormal bases from a matrix A. In essence, it tells us how to transform from one set of these coordinates to the other. If the columns represent points in space (voltages from electrodes), and the rows points in time, then we will transform from one to the other. If the rows and columns were Cartesian x and y coordinates, it makes no difference to the mathematics.

1. There is a marvelous discussion of singular value decomposition geometry in the small chapter entitled "Matrix Times Circle Equals Ellipse" in [ASY97]. We will follow their logic here, but the original is well worth the reader's time.

2. An eigenvector v of A is a vector such that $Av = \lambda v$; that is, eigenvectors are not rotated when operated on by a matrix, but rather are stretched or contracted by the scalar eigenvalue λ.

3. Recall in the chapter 5 exercises that we enforced symmetry numerically in a covariance matrix P by writing in Matlab or Octave Pxx = (Pxx + Pxx')/2.

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If $v_1, \ldots, v_m$ are the eigenvectors of $A^T A$, we define

$$s_i u_i \equiv A v_i$$

where u_i are unit vectors and s_i are scalars. Multiplying on the left by AA^T,

$$s_i AA^T u_i = AA^T A v_i = \lambda_i A v_i = \lambda_i s_i u_i$$

and that the u_i are eigenvectors of AA^T.

In [ASY97] the geometry of Ax is described as *matrix times circle equals ellipse*. The eigenbasis, the vectors that span the circle x, are the v's. The orthonormal eigenbasis that spans the ellipse and is oriented along the semimajor and semiminor axes is the singular value basis. The magnitude of the axes is defined by the eigenvalue s's—the singular values. Figure 7.1 illustrates the geometry of $Av_i = s_i u_i$. In the figure, each $s_i u_i$ represents a scalar eigenvalue s_i times a column u_i. If we combine all of the columns, we have a diagonal matrix of the scalar values times a matrix of the orthonormal column vectors, as $AV = US$. Right multiplying by V^T, we get the form of the singular value decomposition that we will use

$$A = USV^T \tag{7.1}$$

where

$$U^T U = I$$

$$V^T V = I$$

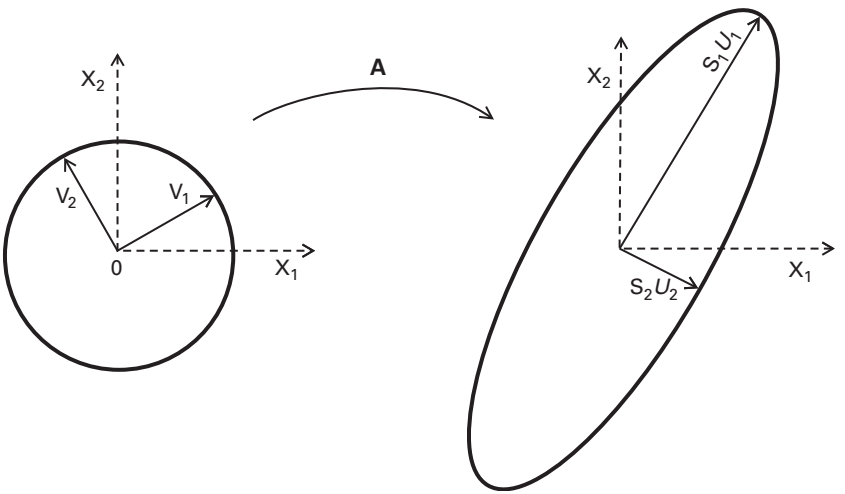


Figure 7.1
The geometry of singular value decomposition $Av_i = s_i u_i$. Reproduced from [ASY97] with permission.

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It is interesting to note that this form of the singular value decomposition is not new. By 1874, Camille Jordan had described this well [Ste93]. By 1907, Erhard Schmidt had described the approximation formulas that we will employ later in this chapter. As with Cholesky's decomposition, described in chapter 2, the mathematics that make the algorithms in this book possible often preceded by decades, or the better part of centuries, our technical ability to incorporate them into digital computers.

One always encounters confusion with comparable techniques in the literature. In *principal components analysis*, the mean of data is often subtracted off, the v_i's are called *components*, and the $s_i u_i$ the *scores*. In practice, principal components are typically calculated from square symmetric matrices, often calculated by forming $A^T A$ or AA^T from rectangular data matrices A. Of course, a strong method of calculating principal components is to use singular value decomposition. Similarly, physicists have a predilection to use Karhunen-Loève decomposition in dealing with discrete data such as we are describing here, although historically this nomenclature began with a focus on infinite dimensional stochastic systems. Psychologists often use *factor analysis*,[4] which is a variation of the principal components (and can be equivalent under certain conditions). Factor analysis arose from the analysis of intelligence, seeking to uncover the fundamental factors responsible for the multifactorial dimensions of intellect. Both mathematicians and physicists have a love of the term *principal orthogonal decomposition*, often used in a deep literature on dynamical systems and coherent structures [HLB96]; it will of tremendous value later in this chapter.

One thing that we need to get out of the way early on is the somewhat touchy subject of the mean—to remove it or not. If you have a cloud of data points that looks like an ellipse, then it might or might not be at the origin, as shown in figure 7.2. If you leave the mean in, the first singular vector or principal component will be along the axis from the origin to the mean. If you are a financial analyst, the market average might be a critical component to track. If you are a neuroscientist, the small changes in activity, riding on top of the ongoing baseline level of cortical activity in an optical imaging experiment, might be the relevant dynamics that are of concern. In biological image analysis, the average level of illumination might have no relevance. On the other hand, if you were studying cortical *steady potentials*, the mean might be a critical component to retain. The message is simple—there is no simple answer. But it is critical to decide whether or not the mean is meaningful in the data to be decomposed.

Let's take a break from single neurons and cortical networks and now examine the dynamics of a network of nine brains.

7.2 The Second Rehnquist Court

The U.S. Supreme Court is typically denoted by the name of the Chief Justice. In what is known as the Second Rehnquist Court, over the eight years from 1994 through 2002, the

4. "Factor analysis is, to put it bluntly, a bitch" (Stephen J. Gould, *The Mismeasure of Man*, 1981 [Gou81]).

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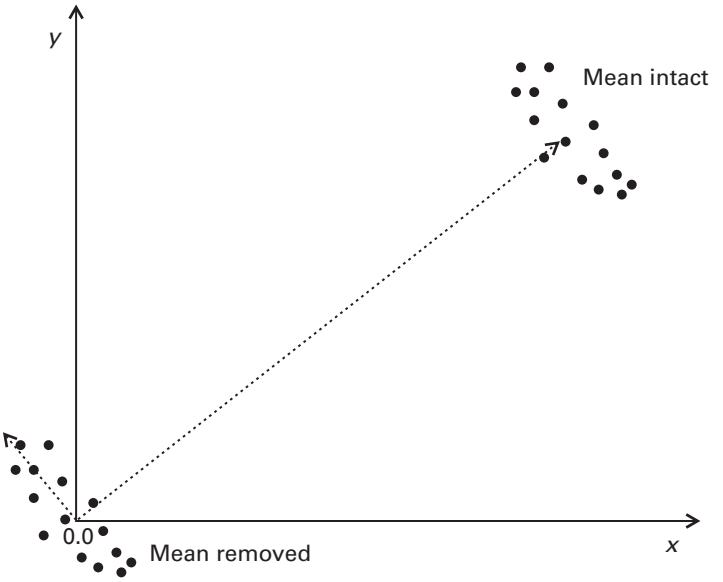


Figure 7.2
Example of the differences in the first singular vector or principal component (gray arrows) based on whether the data mean is left in or taken out.

composition of the nine judges was unchanged (figure 7.3). The court heard almost 500 cases in which all nine judges voted.

The mathematics of singular value decomposition do not care whether we are dealing with images, voltages, or justices. In 2003, Lawrence Sirovich, a colleague who was a household name in the applied mathematics of turbulence and now works on the dynamics of biological systems, published a pattern analysis of the second Rehnquist court using singular value decomposition. Most classroom audiences stop texting (briefly) at this point in a lecture on singular value decomposition mathematics.

The first thing that Sirovich did is to label the justices. The mathematics of singular value decomposition, whether justices or images, do not care about the order in which the data are arranged. So Sirovich used alphabetical order as, well, *justice space*:

R = [Breyer, Ginsburg, Kennedy, O'Connor, Rehnquist, Scalia, Souter, Stevens, Thomas]

Next he creates a *decision vector*, assigning a $+1$ if the justice voted with the majority, or a -1 if the justice voted against the majority. Note that this is a remapping of the true $2^9 = 512$ possible decisions. With a display of mathematical political correctness, Sirovich considers $+$ and $-$ votes to be equivalent, mapping both to $+1$ when they agree with the majority (dissension will be enforced as minority, a -1 vote). This removes a degree of freedom, with $2^{(9-1)} = 256$ possible decisions remaining. This extra degree of

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Figure 7.3
Official group photo of Supreme Court Justices taken in 1998. Photo in public domain.

freedom needs to be accounted for shortly when we calculate the effective dimension of the court.

$n = [n_B, n_G, n_K, n_O, n_R, n_{Sc}, n_S, n_{St}, n_T]$

So if all justices agree, then the decision is *u*nanimous

$u = [1, 1, 1, 1, 1, 1, 1, 1, 1]$

Sirovich gives the example of the vote that helped decide the U.S. *p*residential election in 2000

$p = [-1, -1, 1, 1, 1, 1, -1, -1, 1]$

So there are two ideal models. The *omniscient court*, where they always make unanimous correct decisions, and the justices are clones

$u = [1, 1, 1, 1, 1, 1, 1, 1, 1]$

or the *platonic court*, where each justice sees equally compelling arguments, each justice's decision is as independent as the toss of a fair coin, and there are $2^{(9-1)} = 256$ possible

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Table 7.1
The most frequent patterns of decision of the second Rehnquist court

Br	Gi	Ke	O'C	Re	Sc	So	St	Th	Margin	Frequency % (n)
1	1	1	1	1	1	1	1	1	9	47 (220)
−1	−1	1	1	1	1	−1	−1	1	1	9.6 (45)
1	1	1	1	1	1	1	−1	1	7	4.5 (21)
1	1	−1	1	−1	−1	1	1	−1	1	3.8 (18)
1	1	1	1	1	−1	1	1	−1	5	3.0 (14)
1	1	1	1	−1	−1	1	1	−1	3	2.6 (12)
1	−1	1	1	1	1	1	−1	1	5	2.4 (11)
−1	1	1	1	1	1	1	−1	1	5	1.9 (9)
1	1	1	−1	−1	−1	1	1	−1	1	1.9 (9)
1	1	1	−1	1	−1	1	1	−1	3	1.5 (7)
1	−1	1	1	1	1	−1	−1	1	3	1.3 (6)
1	1	−1	1	1	−1	1	1	−1	3	1.1 (5)

Br, Breyer; Gi, Ginsburg; Ke, Kennedy; O'C, O'Connor; Re, Rehnquist; Sc, Scalia; So, Souter; St, Stevens; Th, Thomas. Data from [Sir03].

decisions. One would get the same dynamics were nine monkeys to be randomly casting votes instead of our idealized Platos.

Using the *information* metric, I, described by Claude Shannon [SW49], one can characterize the novelty of n outcomes as

$$I = -\sum_n p_n \log_2 p_n \tag{7.2}$$

For the omnicient court, there is one outcome, so the $log_2 1 = 0$ and $I_o = 0$ bits of information.[5] For the platonic court, there are 2^8 outcomes, so each $p\, log_2\, p = 1/256 \cdot (-8)$, and the sum yields $I_p = 8$ bits of information. Just as in Shannon's communications information, increasing randomness in a communications channel yields *maximal* information [SW49].

Sirovich estimates a *dimension* $= (I + 1)$ by adding 1 to the information. He notes that out of the 256 possible decisions, 12 account for more than 99% of the decisions, as shown in table 7.1.

Now we ask within what space do the patterns of the justices' decisions live? Using formula (7.2), we add the calculated probabilities of each decision occurring out of the total n, and sum the products of $p_n\, log_2\, p_n$. There are 3.68 bits of information, suggesting that the court acts as if there were only 4.68 instead of 9 justices.

Now let's perform a singular value decomposition to further investigate which justices vote with other justices. We seek a description of whose votes are correlated with whose.

5. A *bit* is the amount of information that you gain from deciding between 0 and 1.

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We stipulate that we will assign these groups in a manner that is independent, and orthogonal to each other. We seek a set of *coherent structures* among the voting patterns. Later in the chapter we will seek coherent structures among patterns of firing neurons rather than among whole voting brains.

First, we set up a data matrix that is structured as shown in figure 7.4. The justices are arranged in columns—they are the *sensors* in our spatiotemporal system (the decisions are, after all, sensing of neural activity within each justice's brain). The rows are the 468 decisions.

Next, Sirovich calculates the first several singular vectors and their singular values as shown in table 7.2. The relative weightings, w_i, are the singular values associated with each vector divided by the sum of all of the singular values, $\lambda_i / \sum_n \lambda_i$. The first thing to notice is that the first two weightings dominate the decomposition. They account for 79% of the variance in the decisions. This means that most of the decision patterns can be expressed by a linear combination of these first two singular vectors. Especially interesting is that we have seen similar vectors before: The first vector is very close to the omniscient decision vector u above, and the second vector is very similar to the presidential decision vector p above. *A linear combination of these two patterns accounted for most of the dynamics of this court.*

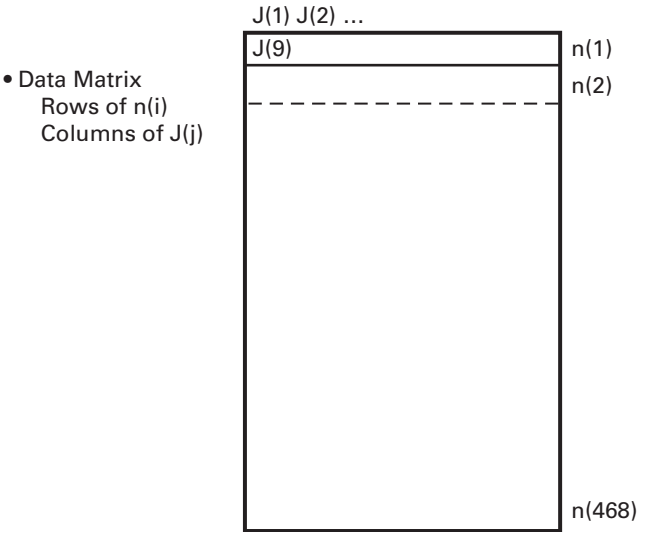


Figure 7.4
Schematic of the data matrix for singular value decomposition. Each decision is represented by a row of nine columns, one for each justice. There are 468 rows for the 468 decisions. Singular value decomposition does not care about the ordering of the justices, but later we will wish to keep the spatial order of neural data ordered so that we can map such data back to their original geometry. For justices, the geometry of their yearly group photos is meaningless.

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Table 7.2
Relative singular values, w_i, and first three singular vectors, v_i, from the singular value decomposition of the justices' decisions

w_1	w_2	w_3
0.57	0.22	0.05
V_1	V_2	V_3
0.34	−0.33	−0.12
0.33	−0.37	0.11
0.36	0.17	−0.35
0.36	0.10	−0.53
0.35	0.30	−0.22
0.31	0.40	0.46
0.35	−0.31	0.08
0.26	−0.44	0.35
0.32	0.41	0.43

Weights $w_i = \lambda_i / \sum_i \lambda_i$. Data from [Sir03].

Last, Sirovich illustrates the decisions plotted graphically, and shows in figure 7.5 how the first two singular vectors span this space. I have sketched into this plot the u and p decisions, which lie very close to these first two vectors. Note that subtracting the mean would give very different results. One would need to consider not the votes, but the deviation of the voting patterns from the average voting patterns. Leave the mean in for a supreme court analysis.

7.3 The Geometry of Singular Value Decomposition

We now need to switch to analyzing spatiotemporal patterns from individual brains. Unlike with the ensemble of justices, we will now need to be kept track of the geometry of the sensor positions. And understanding the geometry of singular value decomposition will be critical.

We go carefully through the geometry of the matrix form of singular value decomposition from equation (7.2) in figure 7.6. The data matrix is A. Each row is an image. If this were data from a two-dimensional electrode array or an optical sensor, then the electrodes or pixels are strung out as a long row. We keep the order of the rows of the data sequentially, so that we can readily reshape the rows back into a two-dimensional matrix for display purposes. But to singular value decomposition the original position within the row does not matter. Time T is indexed in the sequential rows. Sensor number x is indexed in the columns.

Singular value decomposition breaks this matrix up into a U, S, and V matrix. The first matrix U has columns u_i that are the same length of time as the original data. These are

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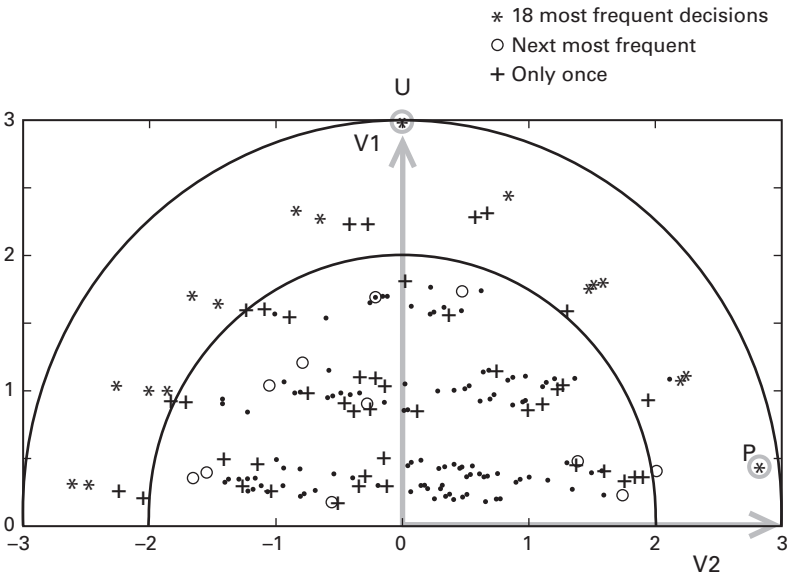


Figure 7.5
Space of decisions, spanned by the first two singular vectors v_1 and v_1. For comparison the unanimous, u, and the presidential, p, decisions are plotted. Modified from [Sir03] with permission.

orthogonal *temporal modes*. On the far right, V is a set of orthogonal *spatial modes* with length x flipped on its side as a transpose. In between is a diagonal matrix of singular values S. U has the dimensions of T x T, V the dimensions of x x x, and S the dimensions of T x x. Note that the number of potentially nonzero singular values are the lesser of T or x to fit on the diagonal within S—the rest of S are zeros. Generally in scientific data collection, you will collect more data, T, than the number of sensors, x. Of course, if you are examining rare events—epidemics perhaps, or phases of the moon—and your data start only when people started recording such phenomena, T might be less than x.

First, I'm going to multiply U by the weightings S. In principal component analysis, this multiplication is the *score*. After each row of U is multiplied against the first column of S, you get a column of $u_1 s_1$. Then, multiplying all the rows of U against the second column of S generates the column $u_2 s_2$, and so on. Thus, each temporal mode becomes weighted by the singular values. If $T > x$, only the first x temporal modes are retained. Now see what happens if we do an outer product between the column $u_1 s_1$ and the first row v_1 of V^T. This generates a matrix with the same dimensions of T rows and x columns. Look carefully at what this gives. The first row in this outer product is literally the first point in time of the first temporal mode weighted by the eigenvalue times the first spatial mode. The second row is the second point in time of the first temporal mode weighted by the same eigenvalue and the same first spatial mode. Every row is the same representation of this eigenimage

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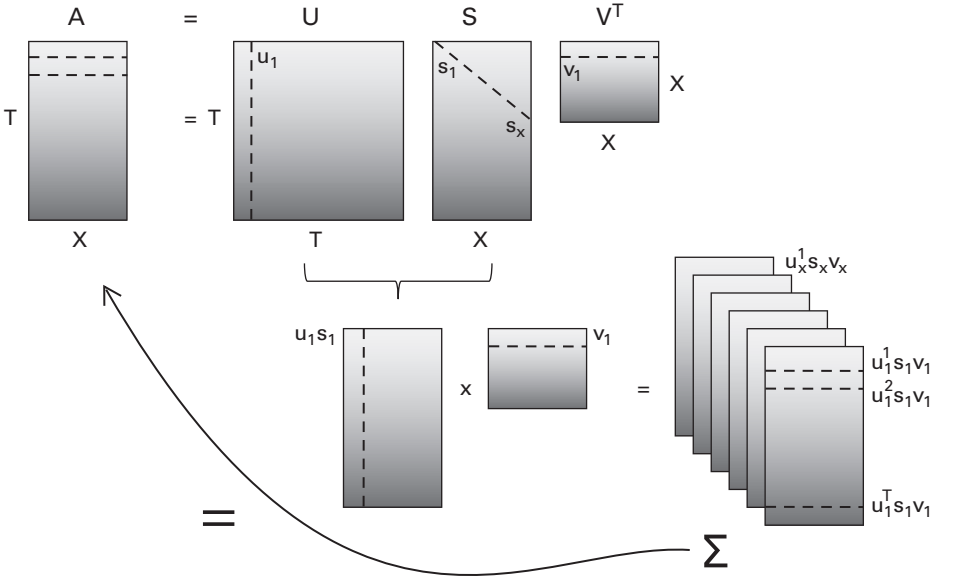


Figure 7.6
The geometry of singular value decomposition.

or eigenvector scaled by the temporal mode. It's a time sequence of the same image if we have two-dimensional data. We will show this in another way shortly.

The second weighted temporal mode, the column $u_2 s_2$, forms an outer product with the second spatial mode, the second row of V^T. We proceed through the x columns of US and form outer products with each of the x rows of V^T. We now have a stack of matrices. Summing this entire stack of the matrices gives us the original A—exactly A. Look at the first row of $u_1^1 s_1 v_1$, and note that there is a stack of rows behind this, that form a linear combination that sums to the first data sample in row 1 of A. This is a linear combination of *all* of the spatial modes. *The data are a linear combination of the modes, and the modes are linear combinations of the data*. The corollary of this is that the modes can contain only what was observed in the data.

What we have done is to find the best representation, on average, for the T images that make up the T rows of matrix A. Let's call this best fit v_1. This first spatial mode maximizes the inner product average of all of the rows, which can be written as

$$\text{maximize } \left\langle (A, v_1)^2 \right\rangle_t = \frac{1}{T} \sum_t (A, v_1)^2$$

where (A, v_1) indicates inner product projection, and $\langle \cdot \rangle_t$ indicates average over time t. There is a remarkable set of relations that emerges among the U, S, and V of the singular

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value decomposition:

$$s_i u_i = (A, v_i)_x$$

$$s_i v_i = (A^T, u_i)_t$$

There is nothing in the u's or v's that was not present in the original data, A.

Now note that the singular values are ordered from large to small, $s_1 \ldots, s_x$. The first singular values are large compared with the later values that are small and eventually meaningless. By the turn of the twentieth century, Schmidt had understood that you could approximate A by using only the top subset of the stack of $u_i s_i v_i$ matrices. This lets you do a crude form of image compression. It also lets you reduce the dimensionality of the space you are working in. Keeping the top two of the nine matrices is what Sirovich did in the Supreme Court example when he showed, in table 7.2, that the top two singular values were much larger than the rest, and figure 7.5 represents a projection of the nine-dimensional justice space in the approximated two-dimensional space following singular value decomposition.

7.4 Static Image Decomposition

Now let's examine a popular image. In figure 7.7 (plate 10), we show the original image, and in the second column, the first four matrices $u_i s_i v_i$. Above the stack of matrices is the partial reconstruction $\sum_{i=1}^{4} u_i s_i v_i$. The image is 124×180 pixels, so there are 124 singular values s_i. If we sum the entire stack to create a full reconstruction, $\sum_{i=1}^{124} u_i s_i v_i$, we get our original image. Now note the modal coordinates, the $s_i u_i$.[6] The first mode is much larger than the others. Why? Because the mean was left in this image. So let's remove the mean, as shown on the far right column. This image looks a bit odd, but the modal coordinates now reveal the details of the image. You can see the belly of the duckie in the first mode, and how the other modes have an orthogonal look and feel from visual inspection compared with the first mode and each other around the belly. Now look at the plots of $log\, s_i$. Comparing these plots with the mean left in with those with the mean removed, you can see (arrows) the substantial difference that the first mode makes with the mean left in. The mean is the average level of illumination of the image. It is important to decide whether this feature is important to the dynamics of image analysis, and whether it is a substantial feature. Note the *tolerance* in the plots of the singular value spectra in figure 7.7. There are many ways of doing this, but in this low-noise figure, I used the product of

6. I am using a terminology that will be more palatable shortly when the u_i values are temporal modes. For now, these are really the modes in the vertical direction of the original image.

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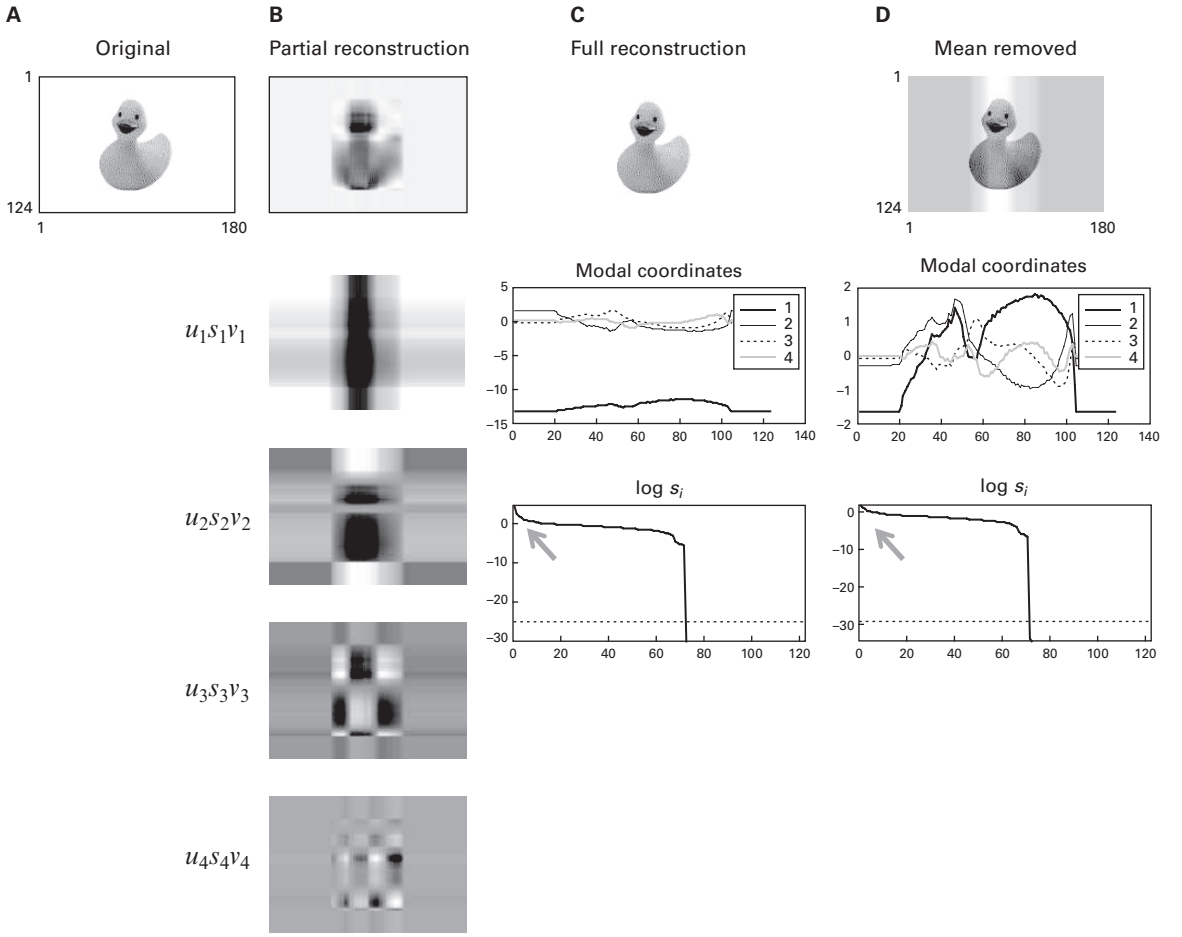


Figure 7.7 (plate 10)
Singular value decomposition of a popular image in A. In B are four matrices $u_i s_i v_i$ and the sum of these matrices is shown at the top. C shows a full reconstruction with the mean retained in the image. D shows the image with the mean removed. The arrows point to the difference that the large first mode contributes to the spectrum of singular values s_i. The horizontal red lines indicate a *tolerance* (red lines) based on the product of the largest of the singular values times the computer machine precision—anything below the tolerance line is computationally meaningless.

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the largest singular value times machine precision of the computer. Anything below this line is computationally meaningless. Regardless of whether the mean is left in or out, it is clear that about half of the $u_i s_i v_i$ matrices can be thrown away—they are multiplied by s's so small that they are meaningless in the reconstruction of the figure.

Next, is the subject of *energy*. If we were studying a dynamical system, such as a velocity field in turbulent fluid, then the square of the singular values would be related to the kinetic energy of the system (through $\frac{1}{2}mv^2$). The static image used here has no intrinsic energy, but we can use the relative sum of the squares of the singular values,

$$\text{Energy} \sim \sum_{i=1}^{m} s_i \Big/ \sum_{i=1}^{n} s_i$$

as a *probability* that the first m out of the total n components account for the energy or variance in the image. For the case when the mean was removed, the first four modes shown account for 90% of the image. When the mean is left in, the first mode, the mean, accounts for 99% of the image, rendering such energy considerations meaningless.

One last thing to consider: The modes are coherent structures. That is, they reveal features that go up and down together. This is why such techniques are useful in financial market analysis—it gives the industries and stocks that are moving together. In our use in neural data, it will show us which neuronal elements are firing together. So let's make this more concrete with an example of dynamic spatiotemporal image analysis.

7.5 Dynamic Spatiotemporal Image Analysis

A very instructive example of the use of singular value decomposition is in the recognition of faces. The so-called rogue's gallery was solved by Sirovich in a series of papers [SK87, KS90], using a database of volunteer students who lent their faces to computational image analysis fame.[7]

In figure 7.8, we show on the lefthand column the first six facial images from this database. The left column in the figure is literally the data matrix if you were to string out each 128×128 pixel image into a single row of 16,384 image intensities. The data matrix is 6 rows by 16,384 columns. Shown on the top row are the six spatial vectors v_i, or eigenimages, reshaped back into their 128×128 locations. The first mode looks like a face, and the others a bit less so. But the first spatial mode is not any of the faces present in the data matrix. It is the image that is the best statistical representation of all of the faces, and computationally would give the largest set of inner products were one to project each of the original faces onto a single face. It is the best linear combination of the original faces such

7. There is an unpublished but widely available 2003 tutorial on image analysis by Sirovich that also employs the rogue gallery, and is highly recommended.

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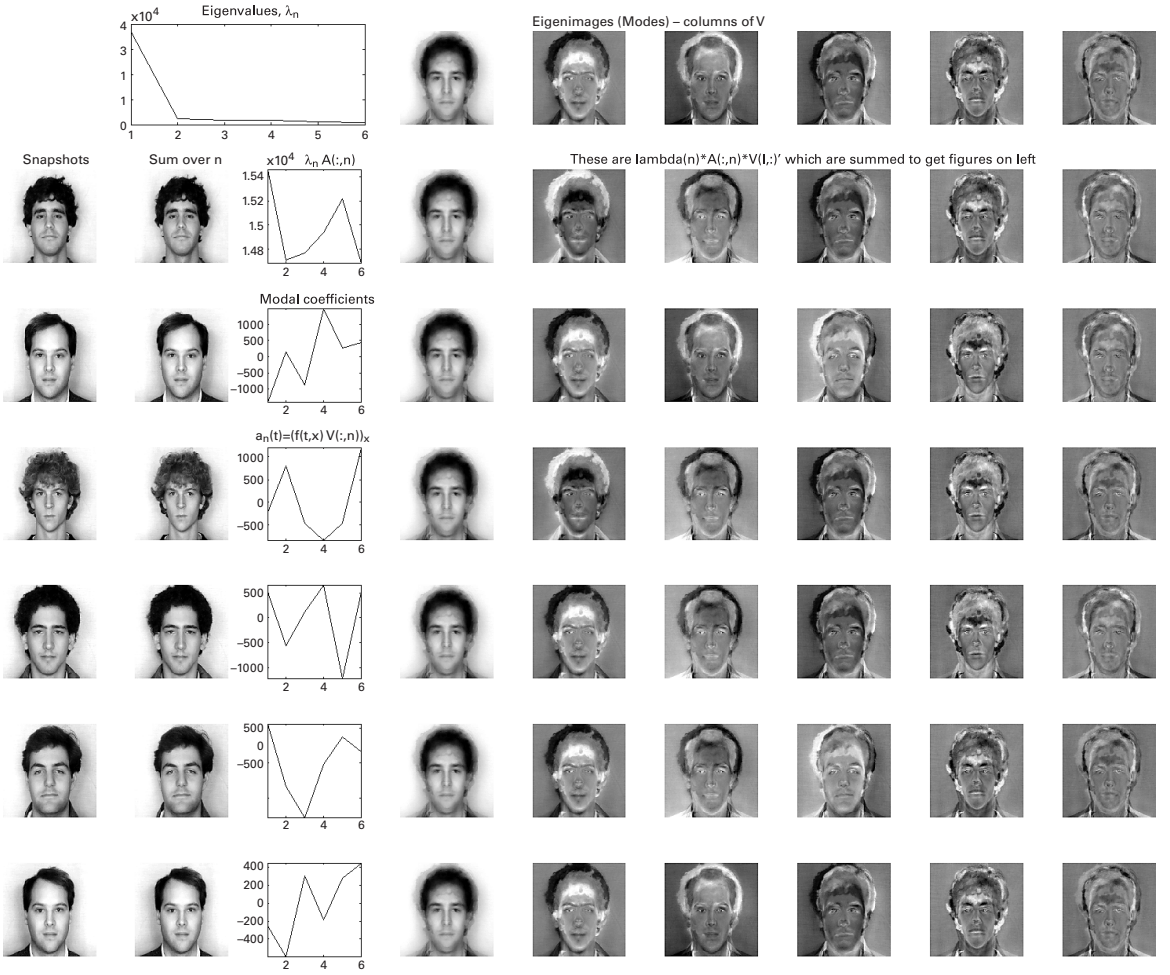


Figure 7.8
The rogues' gallery. Database from [SK87] and [KS90]. The six faces selected are the same as from Sirovich's unpublished tutorial from 2003.

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that these inner product projections are maximized. Each subsequent spatial mode along this top row is orthogonal to the other images. The eigenvalue spectrum is shown rapidly decreasing after the first one—the subsequent modes after the first largely add detail. But, of course, this is the detail that makes each face distinct. In the third column are the temporal modes. Note that the first one is very large (multiplied by 10^4). Forming the outer product of this first temporal mode with the first spatial mode gives the time sequence of the first temporal mode in the fourth column. This is the same as the first $u_i s_i v_i$ matrix on top of the stack of matrices shown in the lower right of figure 7.6. Each image is a subsequent row in this matrix. Forming the outer product of the second temporal mode with the second spatial mode gives the second $u_i s_i v_i$ matrix in this stack, and is the fifth column of figure 7.8. If we sum the entire stack of $u_i s_i v_i$ matrices, each corresponding row sums to the original image in the data matrix, shown as the second column from the left, identical to the first column.

The heuristic value of detailing these faces is that if you understand the geometry, then you completely understand figure 7.9.

7.6 Spatiotemporal Brain Dynamics

Figure 7.9 is a spatiotemporal analysis of voltage-sensitive dye imaging of wave pattern formation in cortex. In this figure, we show data from [HTY+04]. The left column are the first 6 snapshots of the images from a 12×12 photodiode array sampling at 1,000 times per second. There are 1,000 images in the data matrix, recorded over 1 second. I show only the first 6 on the left in order to be homologous with the previous face figure, but the $\cdots$ reminds you that this is just the very upper left corner of a huge set of images. Note that the first and second temporal modes oscillate prominently at 10 cycles per second. If you *filtered* by keeping only the first two modes, you would extract the powerful coherent oscillations at 10 cycles per second.

Now focus on the top row in figure 7.9—the spatial modes. The first spatial mode shows a left-right pattern. The second mode is more diagonally oriented. Although there is a lot of structure in the first four modes, by the fifth and sixth modes, there is considerably less structure. This 1 second of data was taken when the cortex was very coherent, and generating a rotational spiral. It takes only a few modes to account for most of the activity. The brain dynamics here can be expressed in only a few dimensions (modes).

If we sum the corresponding rows of the outer product matrices as discussed earlier, we get the second column. Here, all 1,000 rows sum to reconstruct each individual original image (actually a row in the data matrix). It really is the same as the faces.

You can put anything through such a *proper orthogonal decomposition*—Supreme Court judge decisions, voltage-sensitive dye images, frequencies. So let's do spatial frequencies. We perform a spatial Fourier transform on the snapshots along the left column of figure 7.9. We use wave number instead of frequency in the transform. Let's look at a simple example of this in figure 7.10. We start with a vertical grid pattern. Shifting the zero spatial frequency

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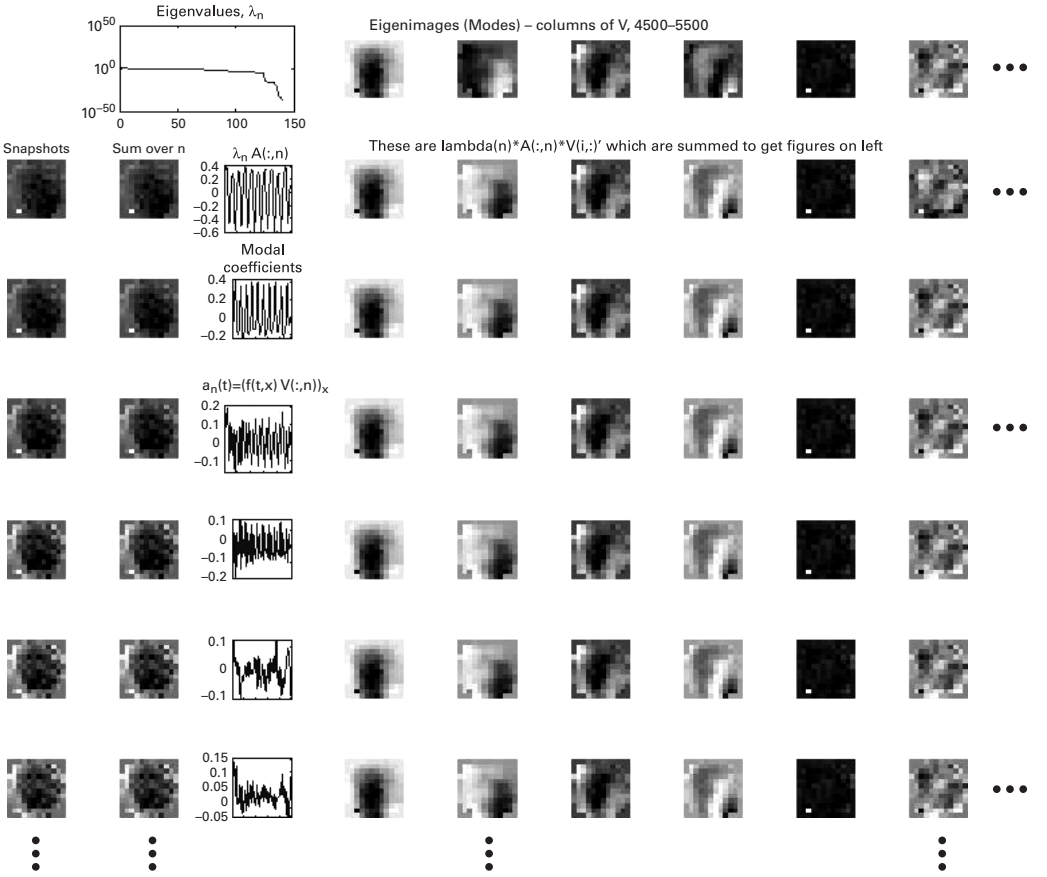


Figure 7.9
Spatiotemporal analysis of brain optical imaging dynamics in the same fashion as shown for faces in figure 7.8.

to the center (crystallographers do this when they present their data), we get a dot at the center, and two dots along the x-axis reflecting the spacing of the grid lines. If I now remove the mean, the point at the center disappears (second column). Now create a checkerboard in the third column. The spatial frequencies form a star pattern, and if I remove the mean in the last column, the center point again disappears. The spatial Fourier transform must be done in the original two-dimensional space and arrangement of the original image (or it would make no sense). But after the transform is done, you can string out the two-dimensional plot of the transform just as was done for the image amplitudes, and perform a principal orthogonal decomposition as we did for justices, faces, and voltages.

One more technical piece before we return to the brain data: Again, the phenomenal contributions of Lawrence Sirovich come to the fore. I just presented a small piece of a set

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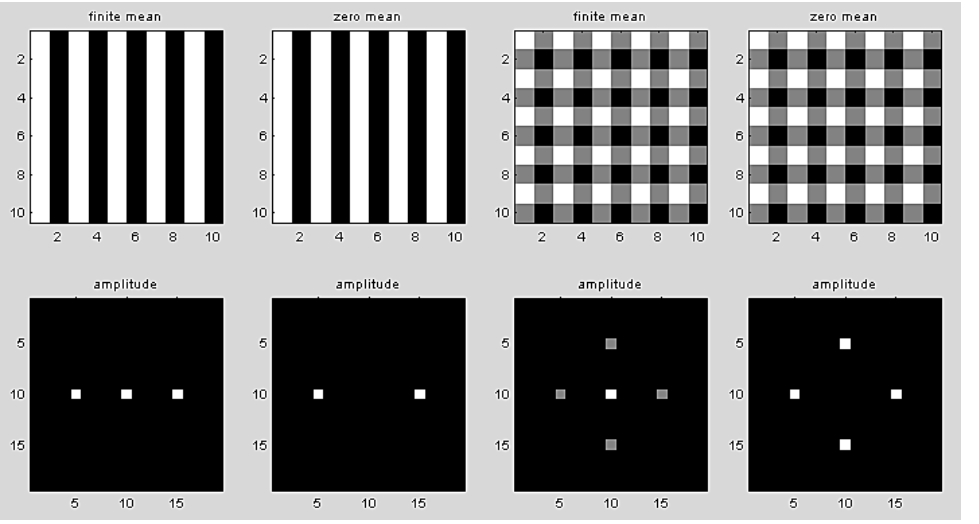


Figure 7.10
Spatial Fourier transform examples.

of optical imaging data—1,000 snapshots over 1 second. But brain dynamics take place over more than 1 second, and each year our imaging cameras, and multielectrode arrays, get larger and faster. Whatever computer you have, at some point you will not be able to solve the singular value decomposition because you run out of memory or your finite lifespan weighs on your patience.

In 1987, Sirovich published a series of three papers in the *Quarterly of Applied Mathematics* on turbulence and the dynamics of coherent structures. In the first of these seminal fluid mechanics papers, he introduced an approach called the *method of snapshots* to deal with the computational problems that large data sets can introduce [Sir87]. For a data matrix, A, where the rows are time, $t = (1, \ldots, T)$, and the columns are space, $x = (1, \ldots, N)$, then for a singular value decomposition,

$$A(t, x) = USV'$$

where $'$ indicates transpose to avoid confusion. U is $T \times T$, S is $T \times N$, and V is $N \times N$. Then we will almost always find that $T \neq N$. For typical laboratory experiments, even for the newest cameras or electrode arrays, $T \gg N$. So we can get rid of T by left multiplying by A':

$$\begin{aligned} A(t, x)'A(t, x) &= VSU'USV' \\ &= VS^2V' \\ A(t, x)'A(t, x)V &= VS^2 \end{aligned}$$

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which is an eigenvalue equation where V is the matrix of eigenvectors of $A'A$ and S^2 is the eigenvalue matrix. And since $N \ll T$, it is a much easier eigenvalue equation to solve now that T is removed. Once V is solved, one can calculate S and then determine U from

$$U = A(t, x)VS^{-1}$$

If $N \gg T$, a comparable solution for U as an eigenvector matrix of AA' can be found.

Using the same data as the detailed study of the dynamics of the cortical wave patterns shown in the previous chapter [HTY+04], we examined the principal orthogonal decomposition of such dynamics in both the amplitude and spatial frequency domain. Figure 7.11 shows a sequence of such dynamics as the cortical network progresses from chaotic, through spiral and plane wave dynamics, to finally dissolution through chaotic disorganization. The first set of segmental eigenmodes corresponds to chaotic activity, the second set to spiral wave dynamics, and the third set to chaotic activity. The modes are plotted both as amplitude and spatial frequency modes. Note the trajectory plots in the third column from the right. These three-dimensional plots are created by plotting the first three temporal modes against each other. On the far right are plotted the global eigenmodes calculated for the entire data sample. These latter calculations are the ones that most require the use of the method of snapshots discussed earlier.

Modes can help in filtering and denoising complex data. In their study of dynamic brain imaging [MP99], Mitra and Peseran show how using the largest modes can denoise such data effectively, as shown in figure 7.12 (plate 11). On the left, they show the average spectra from the first twenty modes, summed much as we summed the largest modes in figure 7.7. On the right side of this figure is a snapshot of our spiral cortex data, below that the spectrogram of the original data set, and below that the spectrogram from a reconstruction of the largest six modes. The evolution of the dynamics can be seen much more clearly in the modal filtered version.

There is a very interesting set of modal decompositions of visually excited turtle cortex from Senseman and colleagues [SR99]. This is a creative work, whose modal trajectories we have adopted successfully in this book. The researchers found something remarkable: The first mode accounts for almost all the dynamics, regardless of whether they use diffuse stimulation with different colors or white light, or spot stimulations. Turtles are not the smartest creature in the pond. But nevertheless, one does speculate whether findings of the large first spatial mode dominating their dynamics, under several conditions, might be related to having left the mean amplitude in their optical images. A reevaluation of such data might prove interesting. But Senseman and colleagues' use of the modal trajectories led to the following.

Look again at the three-dimensional trajectory plots from the modal coefficients in figure 7.11. Such state space reconstructions enable a reconstruction of the images to be performed (the modal coefficients and the eigenvalues determine the weightings on the spatial modes). But they also permit local linear fits to the dynamics in phase space so that a new trajectory,

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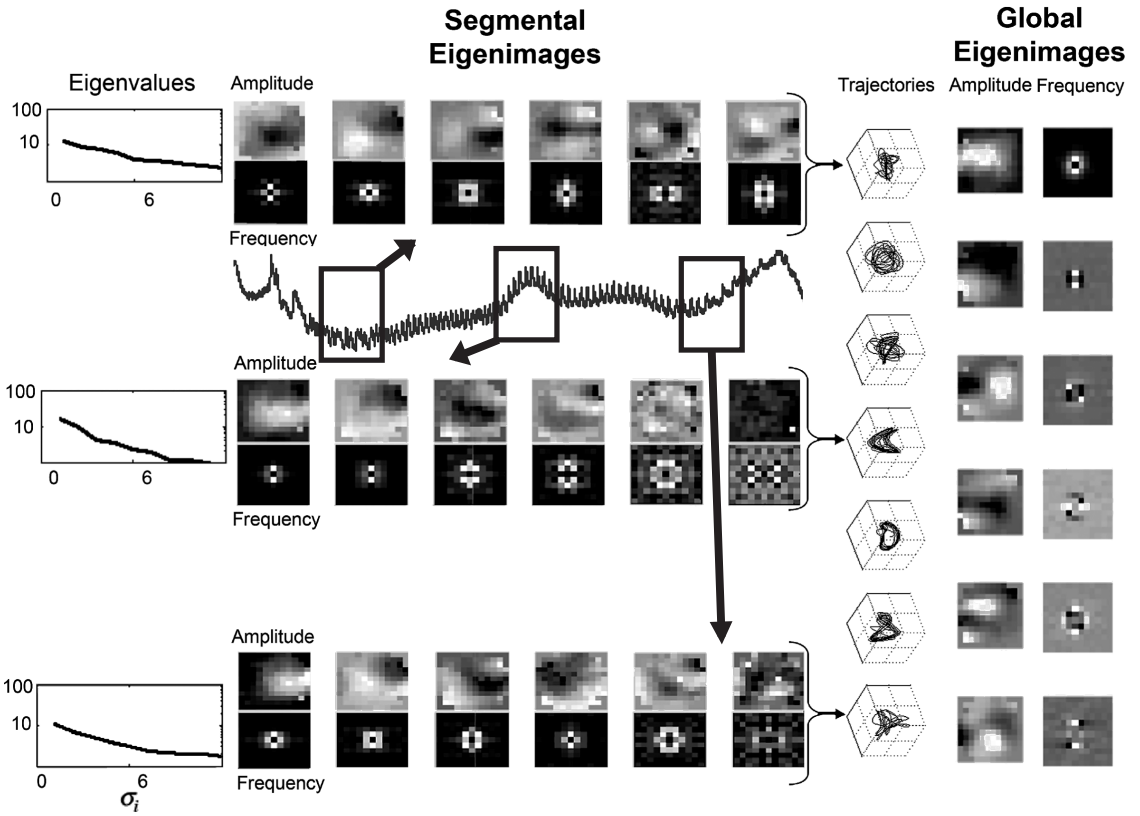


Figure 7.11
Dynamics from a 7-sec oscillatory episode in figure 6.19. At the left side are shown the eigenvalues corresponding to the first six spatial eigenmodes based on the photodetector amplitudes. Below the amplitude eigenmodes are shown the corresponding spatial frequency eigenmodes. These three sets of eigenmodes correspond to seconds 1, 4, and 7 from the episode. The column of trajectory plots from each of the 7-sec episode shows the relationship between the first three temporal modes, u_i. Note the simplification of dynamics during seconds 4–6. On the right side are global amplitude and spatial frequency eigenmodes calculated from all 7,000 images collected. Gray scale proportional to scaled eigenmode amplitudes. Modified from [SHW07] with permission.

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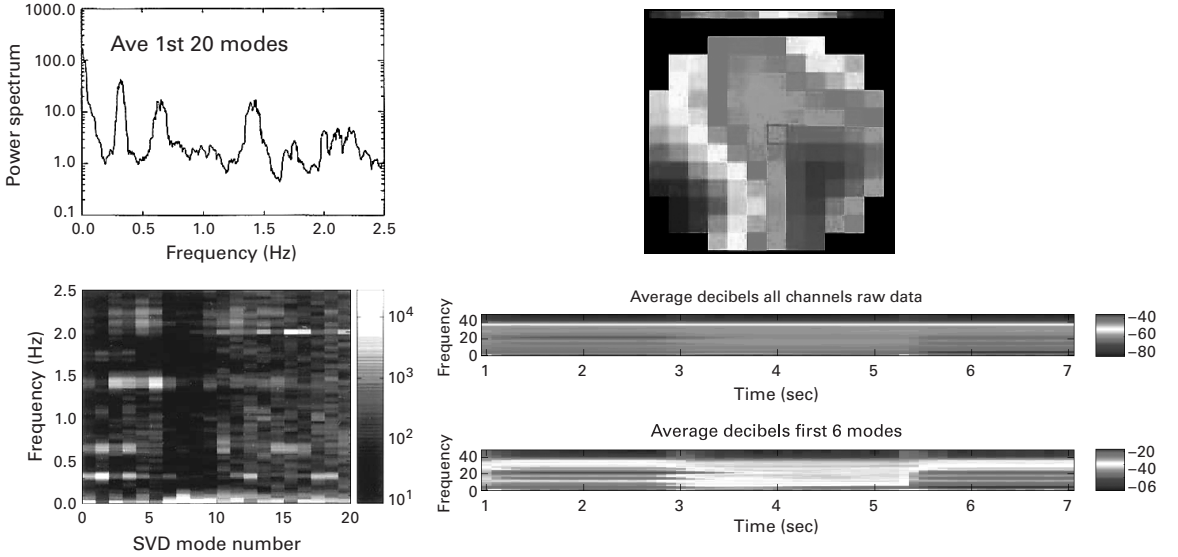


Figure 7.12 (plate 11)
On left is an example of modal filtering with multitapered average spectrum from fMRI dataset, and below that, the spectrogram from individual modes. Reproduced from [MP99] with permission. On the right is an example of our spiral cortical images, with unfiltered and modal-filtered (retention of first six modes) spectrograms. The modal filtered version shows the dynamical evolution much more clearly.

close to previous trajectories, can have a prediction performed based on the nearby trajectories of previous close states. It is this approach that enables a predictive controller to be designed based on such empirical mode decomposition. One can develop an ensemble (unscented) Kalman filter approach by scattering points about such local trajectories and iterating them forward in time. How to choose the modes for the axes in such state space modeling remains an open question. What I show in figure 7.11 are the largest three modes. Picking the best fitting modes in sequence instead of the largest ones is straightforward. This remains an open problem as of this writing.

These dynamics do something that the Wilson-Cowan equations never do. They *evolve*. Wilson-Cowan equations asymptote. Real brains don't asymptote. They do, however, change their activity patterns based on the history of their activity. We are, after all, accommodating learning machines.

If we calculate the equivalent energies of these dynamics, and ask how many modes are required to account for 90% of the energy of the system, we obtain a plot similar to that of figure 7.13. These modes that account for the majority of the energy can be equated to an effective dimension for the dynamics, d_{KL} [Sir89], and we see that those effective dimensions decrease during the episode, then increase. The dimension, or effective complexity of the dynamics, evolves. Later (chapter 12), we will show how this can be accounted for, and it will involve the lower-scale metabolism of ion dynamic such as potassium.

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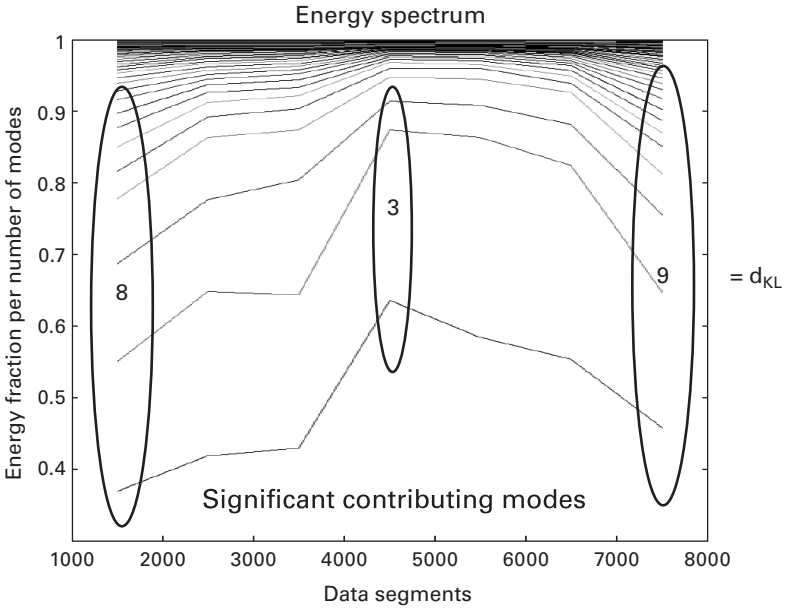


Figure 7.13
Energy spectrum of consecutive data segments from data in figure 7.11. The number of modes required to reach ≥90% of each 1 second's total energy is indicated. Reproduced from [SHW07] with permission.

In figure 7.13, the energy estimates are statistical averages over segments of data. Perhaps one can define an instantaneous effective dimension, D_{eff}, as follows. If the inner product of the data, A, with the spatial modes is (A, v_i), then

$$(A, \mathbf{v}_i)^2 = s_i^2 a_i^2, \quad i = 1...n$$

then we can define $D_{eff} = min(m)$ such that $E(m) \geq 0.9$, where

$$E(m) = \frac{\sum_{i=1}^{m} s_i^2 a_i^2}{\sum_{i=1}^{n} s_i^2 a_i^2}, \quad m < n$$

Instantaneous dimensions are effectively the probability that a given snapshot can be accounted for by a given set of spatial modes [SHW07]. As such, they are practical metrics for real-time control. We can also use them to watch the evolution of snapshot to snapshot, as shown in figure 7.14 (plate 12). The frequency-mode representations, and the frequency-based D_{eff}, are smaller and more compact than amplitude. This makes considerable sense. These waves alternate between rotating and not rotating. If they purely rotated, one would

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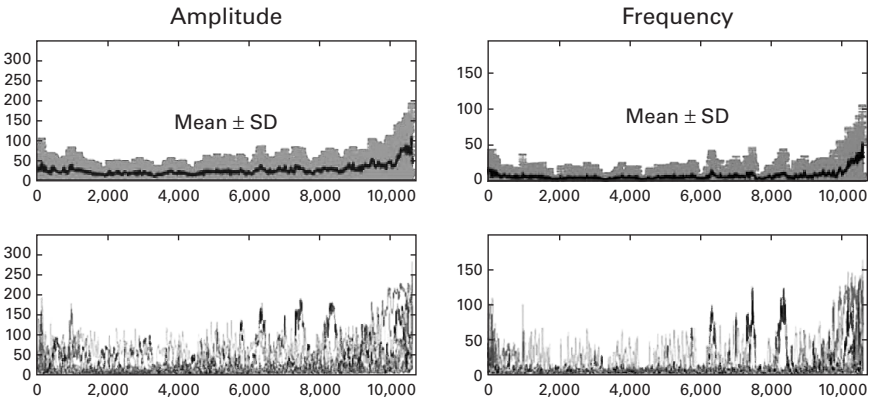


Figure 7.14 (plate 12)
Effective instantaneous dimension shown for each resampled data set, below, and as mean and standard deviation, above, for amplitude and for spatial frequency. Reproduced from [SHW07] with permission.

want to go into a rotating frame of reference that would simplify the representation drastically [BBK03]. An alternative is to use a radially symmetric basis [RFPSS05]. Because we are dealing with brain patterns where the dynamics switch from rotational to irrotational states, I have chosen to stay with a fixed frame of reference. But the reader must keep in mind that there may be better basis representations.

One must be cautious in carrying the analysis from fluid dynamics too far with brain dynamics. Fluids seem simple compared with brains. Their interactions are all nearest neighbor. Brains are nonlocally connected, as reflected in the Wilson-Cowan equations. One technique used frequently in fluid dynamics is to take the basis, such as principal orthogonal decomposition modes, and insert them into the fundamental equations of the fluid dynamics. Then one projects these equations through an inner product onto each basis function mode—a Galerkin projection [HLB96]. For the brain dynamics presented here, the experimental observations suggest that there may be a significant degree of translational symmetry. One observes spirals that drift and wander; they do not seem locked to any particular place in the imaging field. In translationally symmetric cases, a Fourier mode basis may be employed. Especially unclear has been how to handle a Galerkin projection in the setting of the nonlocal connectivity [Lai05] assumed universal in brain network topology. Freestone and colleagues [FAD+11] have recently shown fascinating progress along these lines.

Another issue with brains is that after activity, they are refractory. This means that superposition breaks down. If we employ a linear-mode decomposition technique, then some compensation for the fact that the available brain that can be reexcited changes with system dynamics has not been employed (Tim Lewis, personal communication, 2008). Such a modification of the analyses presented in this chapter would likely be fruitful.

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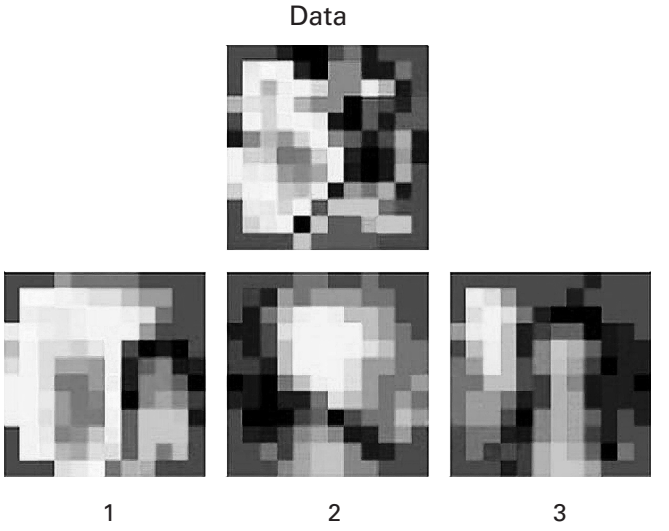


Figure 7.15 (plate 13)
Mode decomposition of a typical snapshot of spiraling cortical dynamics. Reproduced from [SHW07] with permission.

We can project snapshots of raw data onto individual spatial modes, and watch these as animations. Figure 7.15 (plate 13) shows such an example. The raw data are a spiral rotational wave. The first three spatial modes shown are activated in sequence as they pass the rotating wave off from one to another. (An animated movie of these dynamics is archived with [SHW07].) In fluid mechanics one can take a Fourier basis and insert it into the Navier-Stokes equations, or insert into the equations the modes that have been calculated from data. One result is that the modes interact in a peculiar way. These modes are supposed to be independent. But if they really were independent, the modes would just die out. It is the *interaction* between the modes, and in fluid the energy flow between the modes, that keeps turbulence going [SMH05b, SMH05a]. *In the brain, it would be the interaction between the modes, the basic patterns, that keeps brains persistently active.*

The modes from such brain activity are building blocks of activity. Perhaps they are an alphabet, the kind of dynamical alphabet a piece of brain can use in different combinations to produce a variety of dynamics. If so, then the grammar is contained in the dynamics that link the succession of combinations of such modes. In drumheads, for instance, a reasonably small set of prominent vibrational modes, depending on how you hit the drum, recombine to produce the characteristic sound of the drum [Kac66]. The brain, in a sense, is a crystal. It is arranged in a complex lattice, whose connectivity is not just neighbor to neighbor. So brains are a complicated crystal in terms of the nonlocal connectivity. Further work with the types of empirical mode decompositions discussed in this chapter that includes the nonlinear connectivity and nonlinearities of the brain promises a deeper understanding of brain dynamics.

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Do we need to chose between the arrogance of our fundamental models of the brain and the ignorance of our empirical models? In principle, a fundamental model should be more compact for a complex system, and it will always do one thing an empirical model can never do—handle the previously unobserved and unexpected. Fortunately, one can run these different models in parallel. In fact, given the Kalman framework we have presented so far, one can readily compute the prediction errors for each model at each step, and rely more heavily on the one that is more accurate at any point in time. If the computational overhead is practical, running such models in parallel helps us hedge our chances of keeping track of a system that may, or may not, be well accounted for by any one model as it travels through the various regimes of its dynamics.

But for any model, empirical or fundamental, we must formally address their inadequacies. This is rigorously done in the next chapter.

Exercises

7.1. Let's explore singular value decomposition a simple static matrix (figure 7.16):

```
smiley =     [0 0 0 0 0 0 0 0 0
              0 0 1 0 0 0 1 0 0
              0 0 0 0 0 0 0 0 0
              0 1 0 0 1 0 0 1 0
              0 0 1 0 0 0 1 0 0
              0 0 0 1 1 1 0 0 0
              0 0 0 0 0 0 0 0 0
              0 0 0 0 0 0 0 0 0
              0 0 0 0 0 0 0 0 0] ;
```

• Perform a singular value decomposition on this matrix using `[U,S,V] = svd(z);` and extract the singular values from S by `lambda = diag(S)`. Plot the log of the singular values.

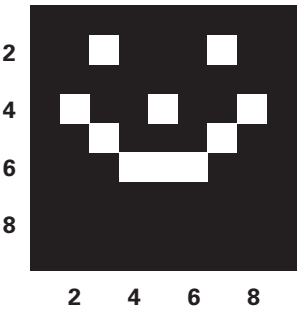


Figure 7.16
Simple smiley matrix.

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Which ones are large enough to be useful? A nice way to guess at the important ones is to estimate a tolerance such as

```
tolerance = max(size(S))*eps(norm(S))*...
                        ones(1,length(lambda));
```

- Now form the partial matrices from figure 7.6 using

```
for i = 1:length(lambda)
  A(:,:,i) = lambda(i) * U(:,i) * V(:,i)';
end
```

Plot these individual matrices. How many do you have to sum in order to get a reasonable approximation to your original smiley face?

7.2. Now let's take a sequence of three images:

```
f(:,:,1) =     [0 0 0 0 0 0 0 0 0
                0 0 1 0 0 0 1 0 0
                0 0 0 0 0 0 0 0 0
                0 1 0 0 1 0 0 1 0
                0 0 1 0 0 0 1 0 0
                0 0 0 1 1 1 0 0 0
                0 0 0 0 0 0 0 0 0
                0 0 0 0 0 0 0 0 0
                0 0 0 0 0 0 0 0 0] ;

f(:,:,2) =     [0 0 0 0 0 0 0 0 0
                0 0 1 0 0 0 1 0 0
                0 0 0 0 0 0 0 0 0
                0 0 0 0 1 0 0 0 0
                0 0 0 0 0 0 0 0 0
                0 1 1 1 1 1 1 1 0
                0 0 0 0 0 0 0 0 0
                0 0 0 0 0 0 0 0 0
                0 0 0 0 0 0 0 0 0] ;

f(:,:,3) =     [0 0 0 0 0 0 0 0 0
                0 0 1 0 0 0 1 0 0
                0 0 0 0 0 0 0 0 0
                0 0 0 0 1 0 0 0 0
                0 0 0 0 0 0 0 0 0
                0 0 0 1 1 1 0 0 0
                0 0 1 0 0 0 1 0 0
                0 1 0 0 0 0 0 1 0
                0 0 0 0 0 0 0 0 0] ;
```

which represent the images in figure 7.17.

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Figure 7.17
Sequence of three simple images.

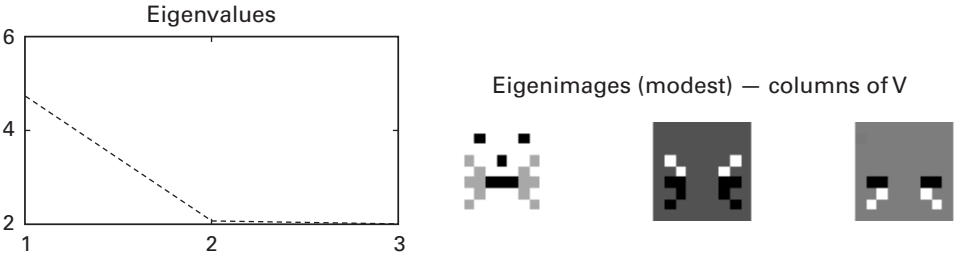


Figure 7.18
Eigenvalue spectra and the three spatial modes.

• First, let's make a data matrix out of these images. Do this by stringing out the images as a set of row vectors, and consider each image (row) as a sequential point in time. One way to do this is to write:

```
[rows,cols, NumberOfSnapshots] = size(f)
for i = 1: NumberOfSnapshots
  ff(i,:) = double( reshape(f(:,:,i),1,[]) );
end
f=ff;
```

• Now you can perform your singular value decomposition exactly as you did earlier with the static image in exercise 7.1, and form the partial reconstruction matrices.

• Plot the eigenvalue spectra, and plot the three spatial modes. You should get something that looks like figure 7.18.

• Plot the three temporal modes.

• Now plot the outer products of the temporal modes and the spatial modes. Each row of the summed partial matrices should sum to give you each of the original three images.

• But more subtly, note how the values of the temporal modes determine how each of the spatial modes is weighted to form the various forms of the smile or frown in these images. You should get something that looks like figure 7.19.

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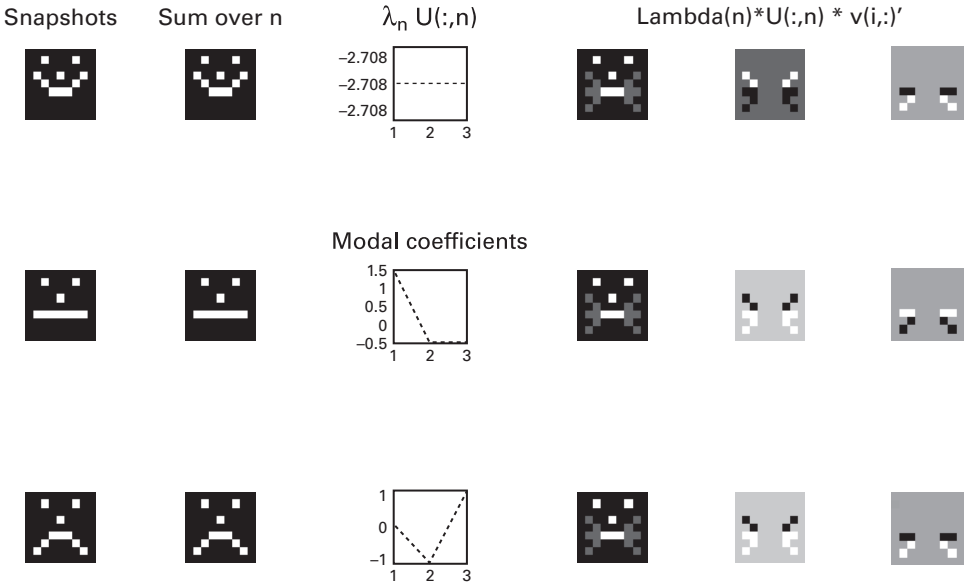


Figure 7.19
Singular value decomposition.

7.3. The canned routine `svd` in Matlab or Octave is great for small data sets, and miserable for larger ones. I will now give you a routine that is very useful for neuronal data sets. It is the code required for solving Sirovich's *method of snapshots* [Sir89], and I will present it here with sufficient annotation so that you can follow the logic:

```
%Test to see which Dimension is the largest
    [T,N] = size(f);
    if N >= T
      %ff' = (ASV')(ASV')' = (ASV')(VS'U') = AS^2A'
      %ff'U = AS^2
      %and the size of ff' is only the length
      %of the number of snapshots ^2
      [U,SS] = eig(f*f','nobalance');
      S = sqrt(SS);
      lambda = diag( real(S) );
      [lambda,IX] = sort(lambda,'descend');
      lambdamatrix(:,1) = lambda;
      S = diag(lambda);
      %Only retain significant eigenvalues
      Retain = length(find(S > 10*sqrt(eps)));
      LeftoverS = S(1:Retain,1:Retain);
      S = LeftoverS;
```

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```
    for i = 1:Retain
      Atemp(:,i)=U(:,IX(i)); %reorders lambda, U
    end
    U = Atemp;
    V = (inv(S) * U' * f)';
  else%N > T
    %f'f = (ASV')'(ASV') = (SV')'U'(ASV') = VS'U'ASV'
    %f'fV = VS^2
    [V,SS] = eig(f'*f,'nobalance');
    S = sqrt(SS);
    lambda = diag( real(S) );
    [lambda,IX] = sort(lambda,'descend');
    lambdamatrix(:,1) = lambda;
    S = diag(lambda);
    %f = ASV'
    %S^- 1A^-1f = V'
    %if f is TxN dimensional, and N >> T,
    %then only first T eigenvalues
    %of S, and first T rows of V' (cols of V) are significant
    %Only retain significant eigenvalues
    Retain = length(find(S > 10*sqrt(eps)));
    LeftoverS = S(1:Retain,1:Retain);
    S = LeftoverS;
    for i = 1:Retain
      Vtemp(:,i)=V(:,IX(i)); %reorders lambda and V
    end
    V = Vtemp;
    clear Vtemp
    U = ( f * V * inv(S) );
  end%
```

This algorithm reduces the problem to the smallest dimension in the data matrix, and allows you to handle much larger data sets for a given amount of computer memory. It can also be much faster even when you have sufficient computer memory to handle the `svd` decomposition directly.

- Use this method of snapshot algorithm to calculate the singular value decomposition, and see whether the results are the same as before.

7.4. You can perform the same calculation in spatial frequency space. To do this and center zero frequency on the origin, consider the following lines of code:

```
[rows, cols, page] = size(f);
spatialft_f = zeros(rows,cols, page); %preallocate memory
for page = 1:max(page);
  demeaned_f(:,:,page) = f(: ,: ,page) - mean(mean(f(: ,: ,page)));
  spatialft_f(:,:,page) = abs(fftshift(fft2(demeaned_f(:,:,page))));
```

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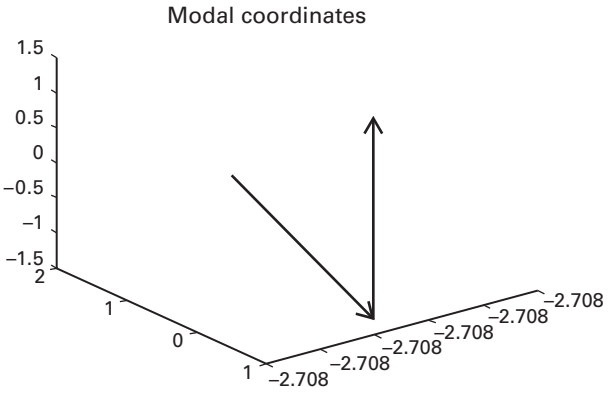


Figure 7.20
Trajectories of temporal modes.

```
end % page
f = spatialft_f; %do in K space
```

You need to do the spatial Fourier transform in the arrangement of the original smiley face images. Then you can reshape the transformed data into the data matrix as done in exercise 7.2, and perform singular value decomposition on this matrix. Examine the nature of the spatially transformed smiley face images. Can you see the differences in structure expressed in the spatial frequencies?

7.5. Last, let's explore the trajectories of the temporal modes plotted in the coordinates of the singular value decomposition. Take the first value of each of the three modes and plot this as a point in space, and then link it to the second, and the third. These are actually the original temporal data transformed by the *matrix times circle equals ellipse* into the space of the new coordinate system in v. In figure 7.11, such a trajectory formed the plots that resembled limit cycles and horseshoes when the dynamics were plane waves or spirals, respectively. For this simple example, your results look like the plot in figure 7.20. But if you had data that exhibited low-dimensional dynamics, and if the patterns recurred, then a local fit to nearby points would give you a predictive model for the next set of values. This is how you can use such empirical models for tracking and prediction.

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